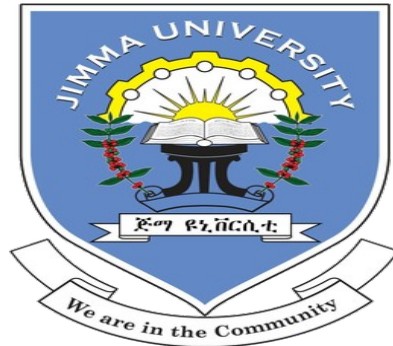


Biochemistry of Cardiovascular System For Pc-Medical Students

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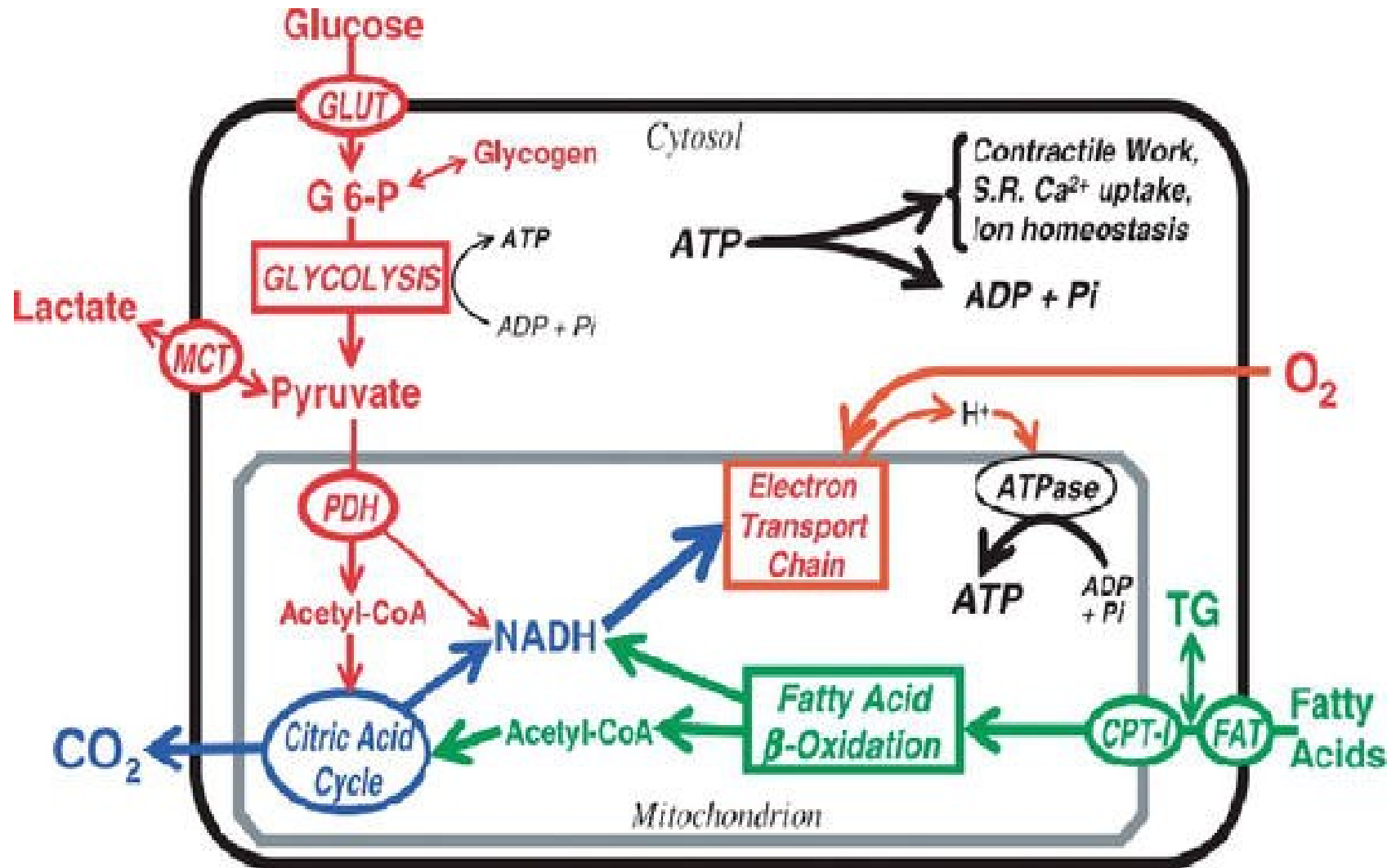


Learning objectives

At the end of this sub-topic, the students will be able to

- Describe the biochemistry of Cardiacenergetics
- Discuss the features of metabolism of cardiac Muscle
- List the major substrates used for cardiac muscles
- Discuss metabolic regulation in cardiomyocyte and effect of ischemic in cardiac metabolism
- Distinguish cardiac biomarkers and their clinical significance

Overview of Coupled Metabolic Rxn in cardiomyocyte



Cardiac Metabolism



■ Cardiac metabolism

- **Producing and using chemical energy (ATP) to power the heart function**
- **Demands about 6kgs of ATP/day**
 - ✓ 70% of energy are used for contraction
 - ✓ The rest for ion pump function
- also involves the **expression of various transporters** role in regulating substrate uptake and energy production and xenobiotics homeostatic machinery.
- Low [ATP] and Many ATP hydrolysis observed

Cardiac Metabolism cont.



- Cardiac muscle adapted to be highly resistance to fatigue
 - Large amount of **myoglobin** and **mitochondria**
 - Continues aerobic ATP production via OP
 - The heart is tuned to aerobic that it is unable to pump in **ischemic condition**
- Normally about 1% energy derived from **anaerobic metabolism** that can increase to 10% under **moderately hypoxia**

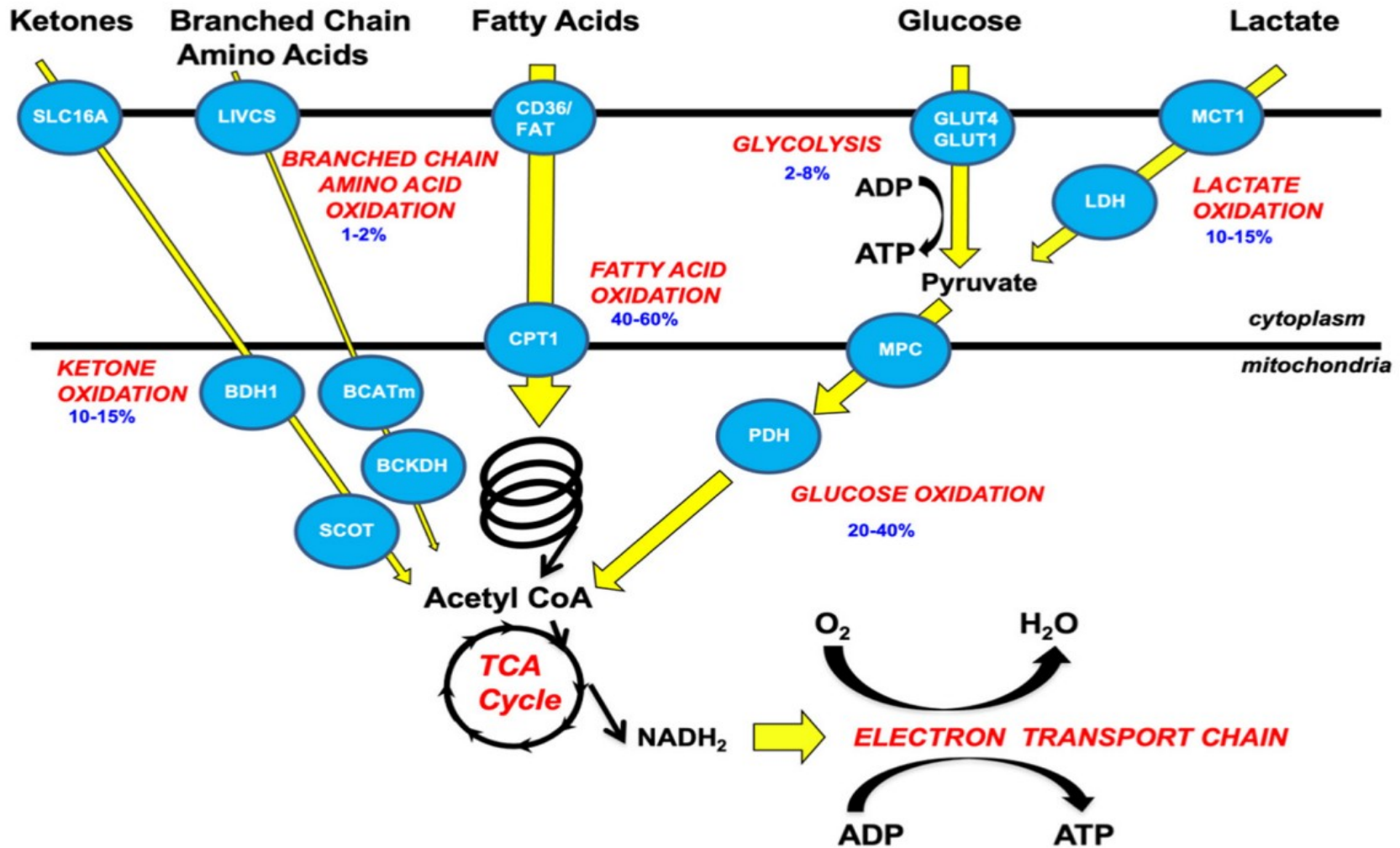
Cardiac Metabolism cont.

- Deregulation of cardiac metabolism
 - ✓ common diseases that leads to heart failure(HF)
- Aerobic production of ATP requires for
 - Na /K –ATPase
 - Counteraction and Relaxation
 - Therefore heart relaxation and contractility leads to increase myocardial metabolism
 - Heart has limited ability for anaerobic metabolism

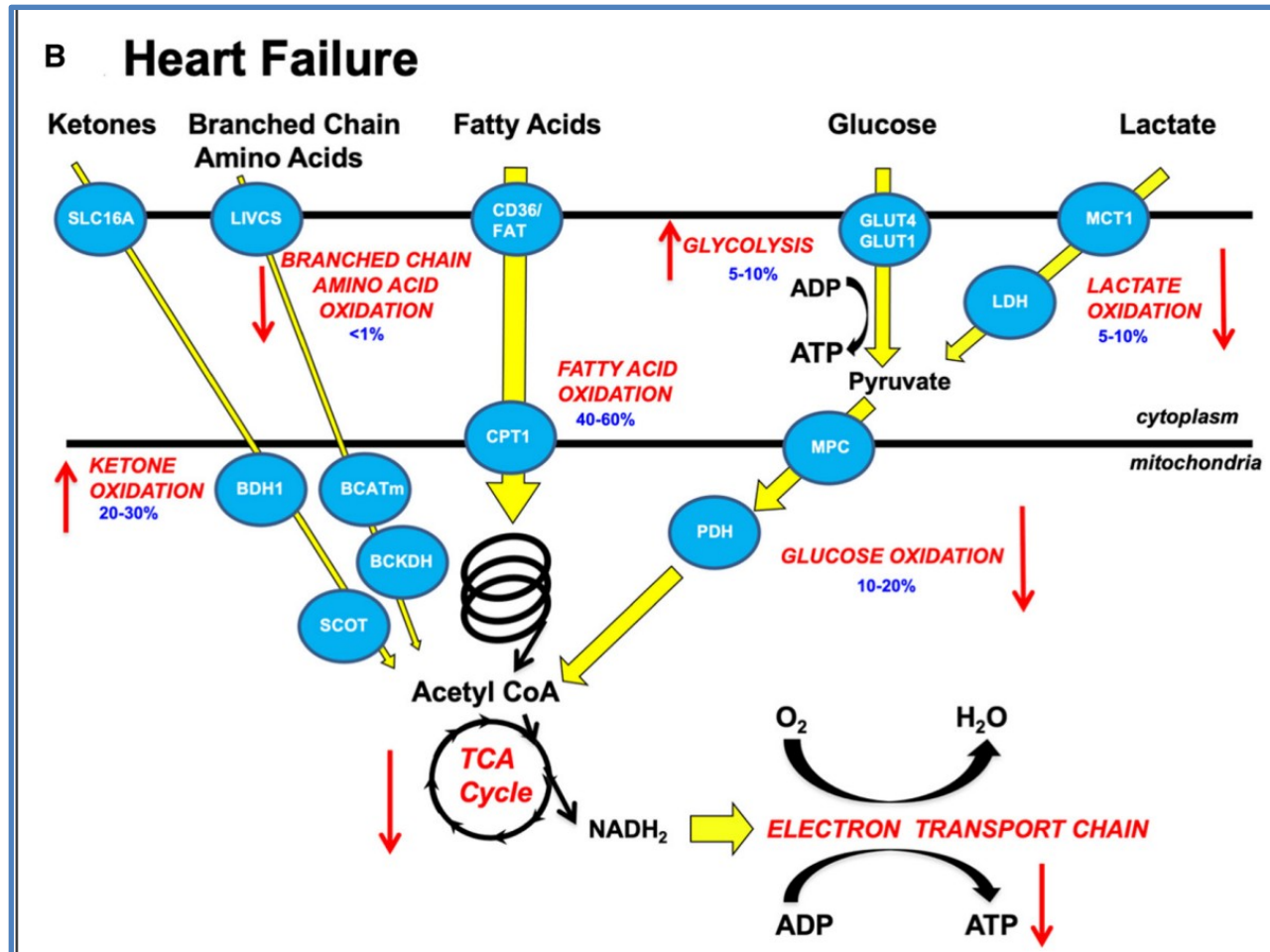
Normal Heart Metabolism



A Normal Heart



Metabolism in Heart Failure



Cardiac Energetics



- The energy requirements of the heart are very high, requiring the equivalent of 6 kg ATP/day.
 - The heart must contract continuously.
- Because the **ATP storage** within the cardiomyocyte is minimal,
 - a tight coupling of ATP **production** and **myocardial contraction** is essential for normal cardiac function.
 - It contains negligible energy reserves in the form of **glycogen or lipid**

Cardiac Energetics Cont.

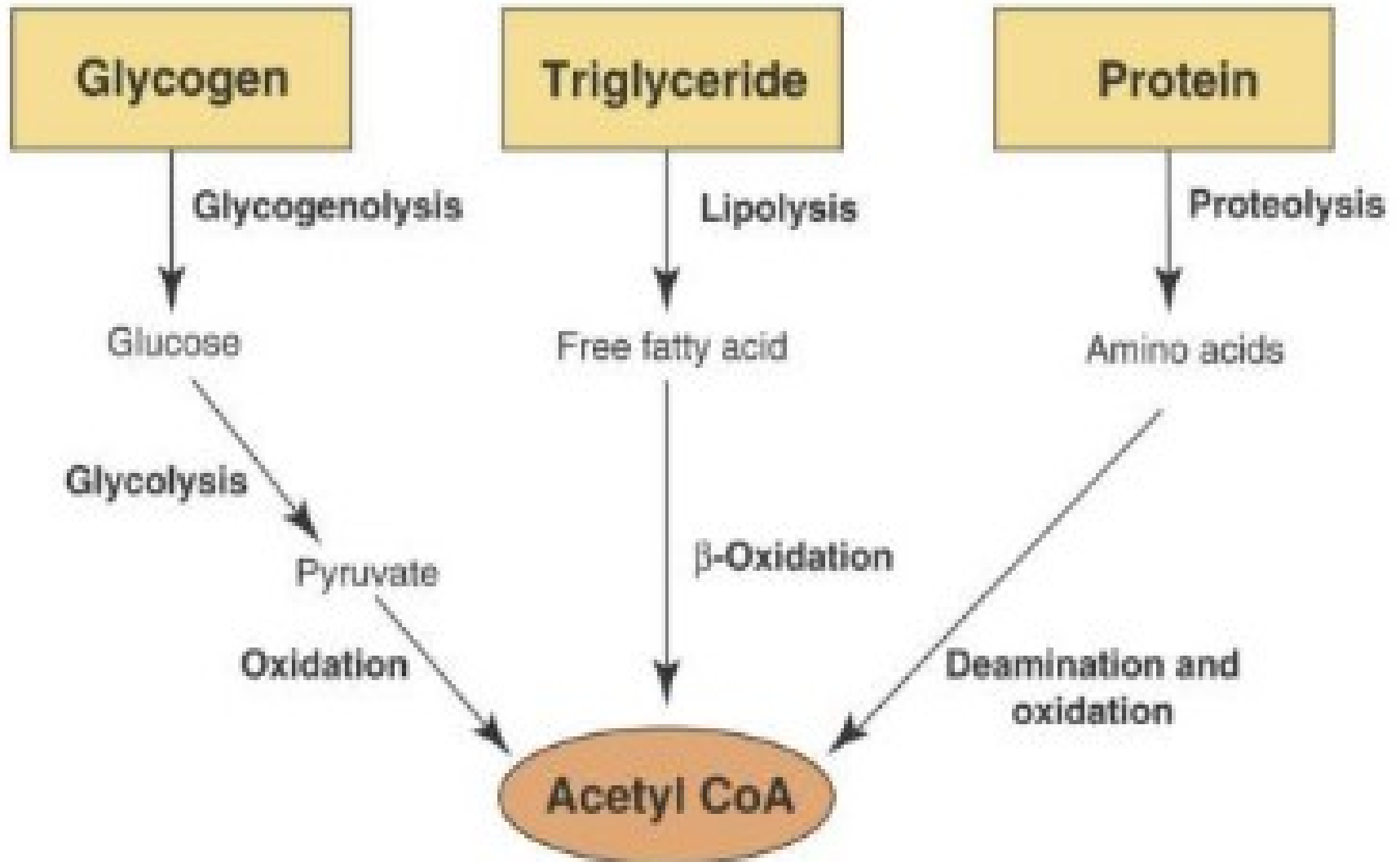
- Cardiac mitochondrion generates **>95% of ATP** used by the heart,
 - Also regulates **intracellular calcium homeostasis**, signaling and cell death.
 - Thus heart muscle, except for short periods of extreme exertion, **relies entirely on aerobic metabolism**
- Therefore, cardiac myocytes is richly with **mitochondria**; they comprise up to 40-50% of its cytoplasmic space.

Substrate Metabolism In The Heart



- Substrates for ATP Production in heart
 - Carbohydrates(Glucose)
 - lipids(Fatty acids)
 - amino acids(BCAA)
 - ketone bodies(BD)
- The robustness of **cardiac metabolism** is reflected by
 - its **highest oxygen consumption rate** on the per unit weight basis.
- For a human heart, the amount of ATP turned over during a 1-day period is 15 to 20 times of its own weight.

Overall pathway



Substrate Preferences



- **Sufficient oxygen**
 - Fatty acids(50-70%)
 - Glucose (30%)
 - Also under ischemic conditions
- **Increased muscular activity**
 - Glucose
 - lactate
- **Pathological condition and starvation**
 - Ketone bodies
 - amino acids

Substrate Preferences cont..



- **Fatty acids**

- the **resting heart's fuel of choice**
- But, upon the imposition of **heavy work load** the heart greatly increases its rate of consumption of Glucose
- However, it is a **metabolic omnivore**, changing according to
 - metabolic demand
 - oxygen and
 - substrate availability
- **crucial for the maintenance of normal cardiac function.**

Substrate Preferences cont..



- **Acute adaptations** (e.g., to increased workload) occur via
 - altered activity of metabolic enzymes

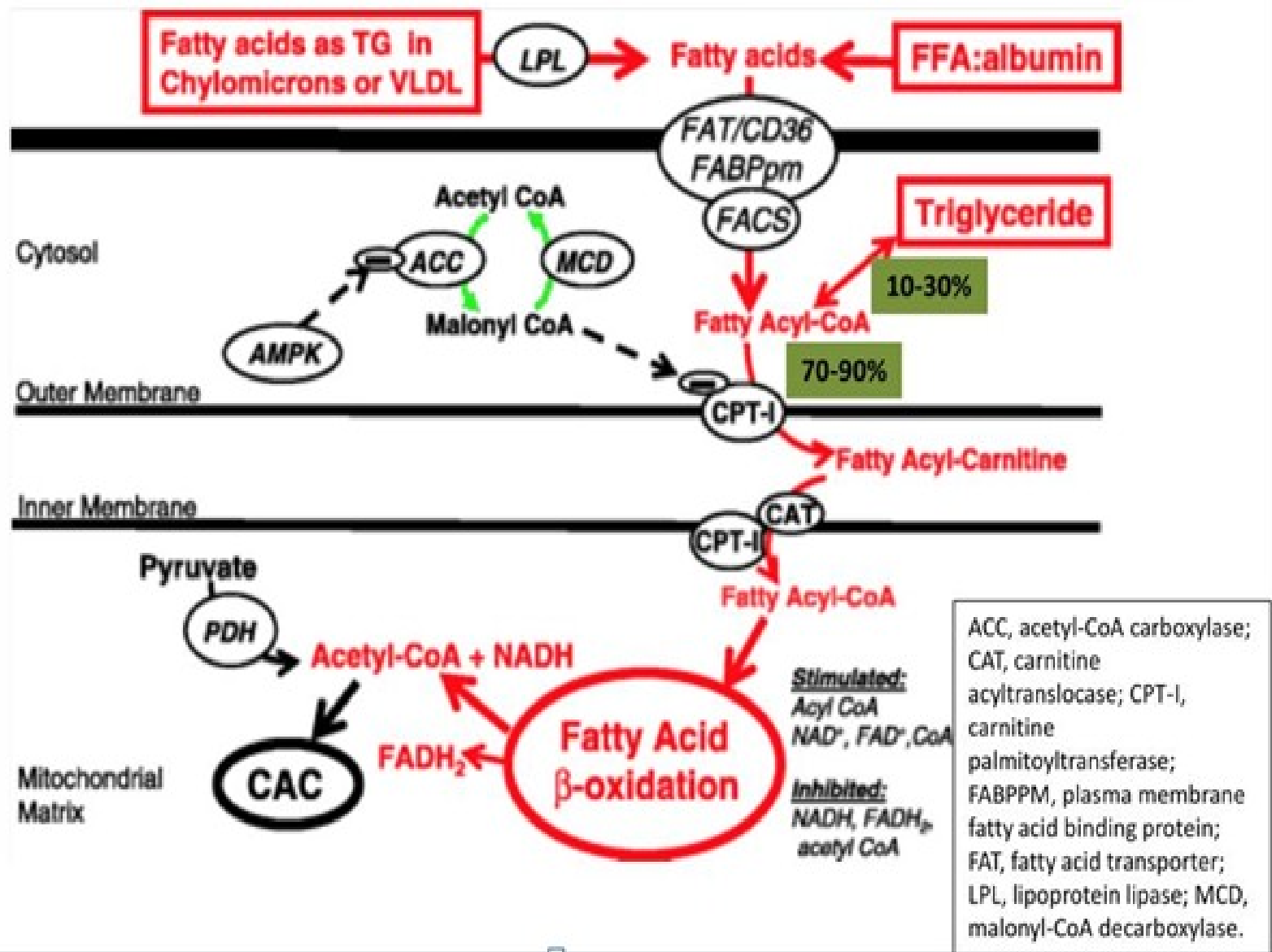
- **Chronic adaptations** also occur in both health and disease related to **altered gene expression**

Substrate Metabolism of Heart

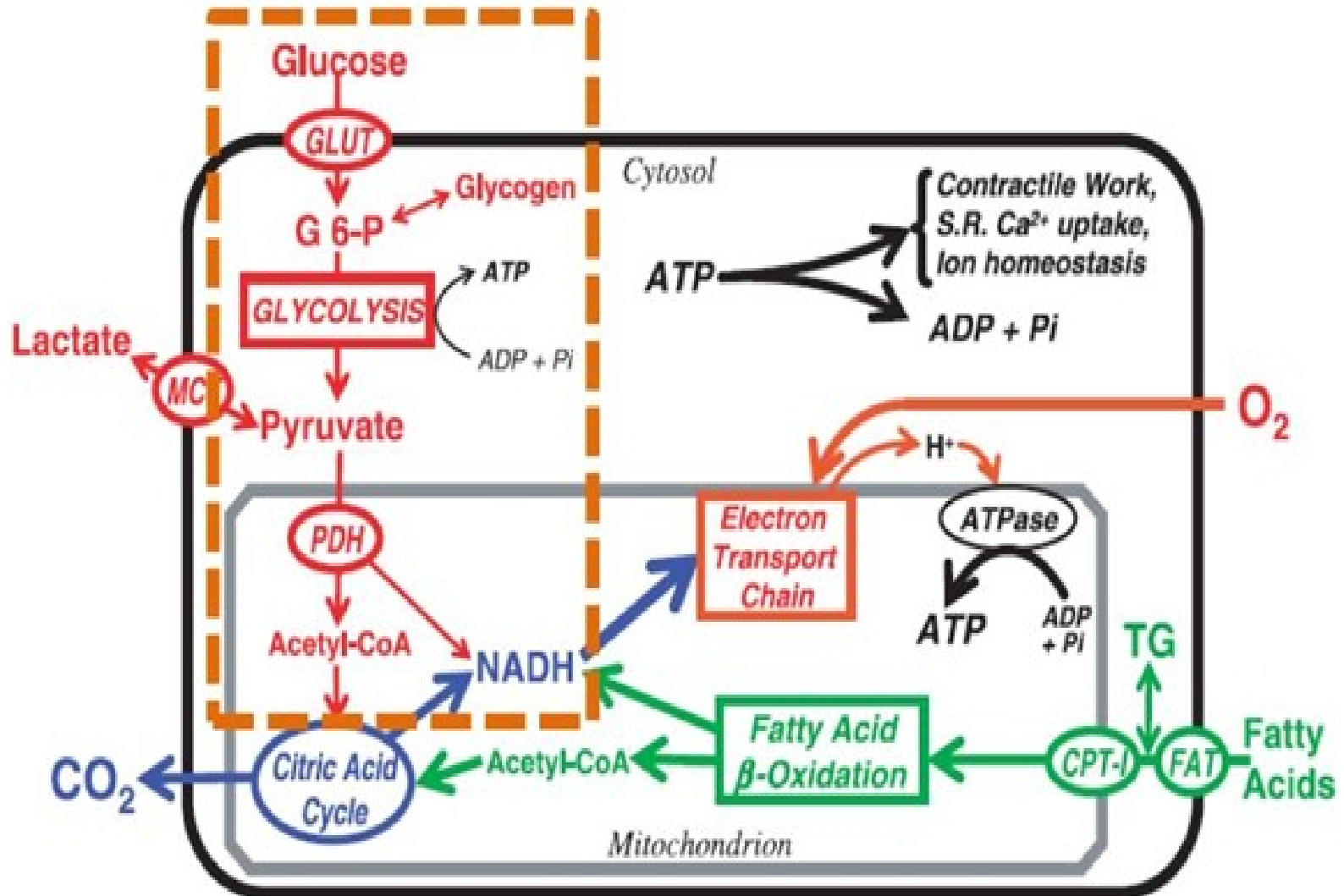


- In a **normal heart**, mitochondria are largely fueled by
 - ✓ **fatty acyl-coenzyme A** - FA
 - ✓ **pyruvate**- CHO
- The entry of long chain acyl-CoA into the mitochondrion is with the **rate-limiting step at the (CPT1)** reaction.
- The oxidation of pyruvate is regulated at the **pyruvate dehydrogenase (PDH) reaction**

Fatty acid Metabolism



Glucose Metabolism



Glucose Metabolism Cont.



- In cytosol, glycolysis
 - glucose is converted to pyruvate
 - PDH enzyme complex in the mitochondrial matrix catalyzes
 - The conversion of pyruvate to acetyl coenzyme A (CoA).
- PDH is a rate-limiting step in COH oxidation
 - being negatively regulated by kinase and positively regulated by phosphatase enzymes
 - Allosteric regulation
 - Positive (upstream, by pyruvate) and
 - Negative (downstream, by acetyl CoA/CoA, ATP/ADP, and (NADH+ H+)/NAD+ ratios)

Glucose Metabolism Cont..



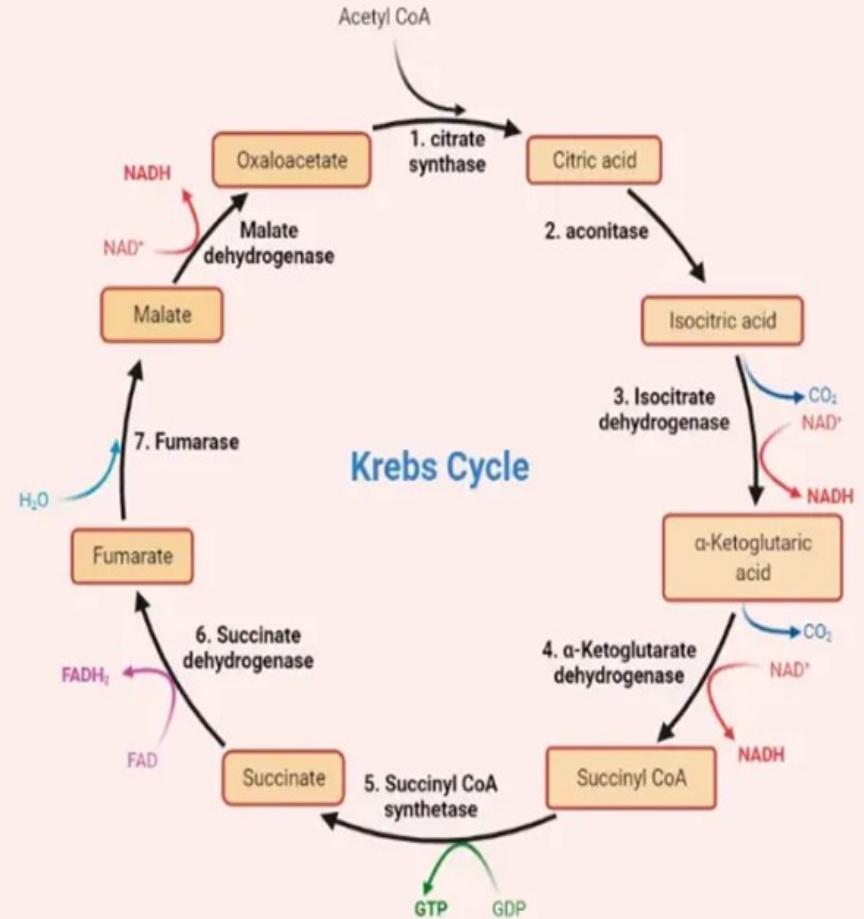
- Hypoxia inhibits PDH via **hypoxia-inducible factor stabilization**, therefore
 - pyruvate is converted to lactate by the enzyme LDH
- Hypoxia also has more generalized effects **on cardiac metabolism** such as including
 - modulating the enzyme activities of Krebs cycle, and
 - controlling the mitochondrial permeability transition pore opening.

<https://www.ahajournals.org/doi/pdf/10.1161/CIRCGENETICS.114.000221#:~:text=Hypoxia%20also%20has%20more%20generalized%20effects%20on%20cardiac,the%20mitochondrial%20permeability%20transition%20pore%20opening%20%28Figure%201%29.http%20s%3A%2F%2Fwww.researchgate.net/publication/313264480> Stage-Specific Effects of Hypoxia on

Glucose Metabolism Cont..



- Under aerobic conditions, acetyl CoA enters the TCA cycle in the mitochondrial matrix.
- One unit of acetyl CoA generate
 - 3 NADH &
 - FADH₂
 - electron donor to complexes of the ETC to produce ATP

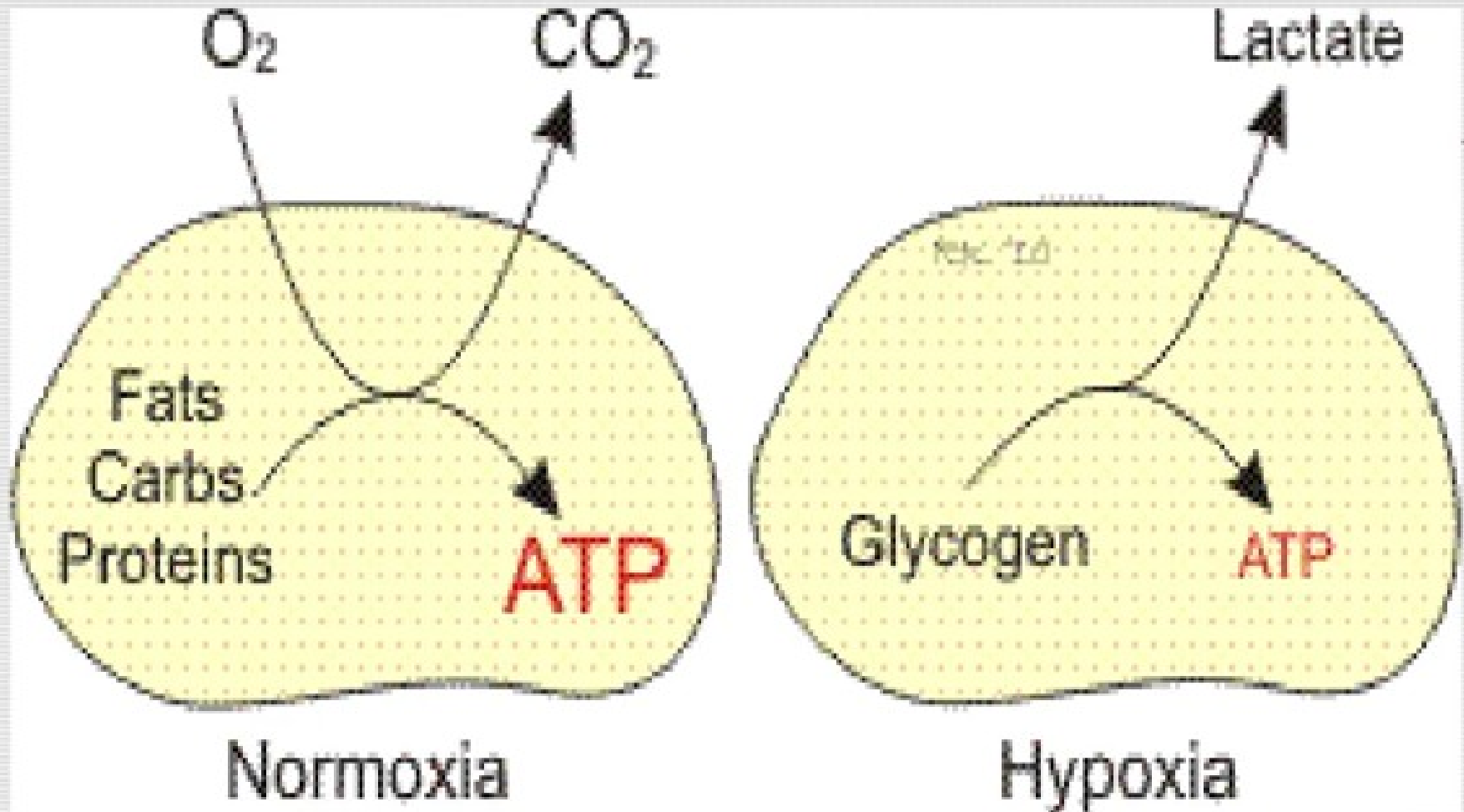




lactate, ketone bodies, and AAs,

- Those substrates
 - can enter mitochondria directly for oxidation
- Metabolism of
 - ketone bodies yields acetyl-CoA, whereas
 - amino acid yields keto-acids
- The contribution of ketone bodies and AAs to overall cardiac oxidative metabolism is considered to be **minor** because of
 - the low availability of these substrates under **normal physiological conditions.**

Substrate Metabolism of Heart

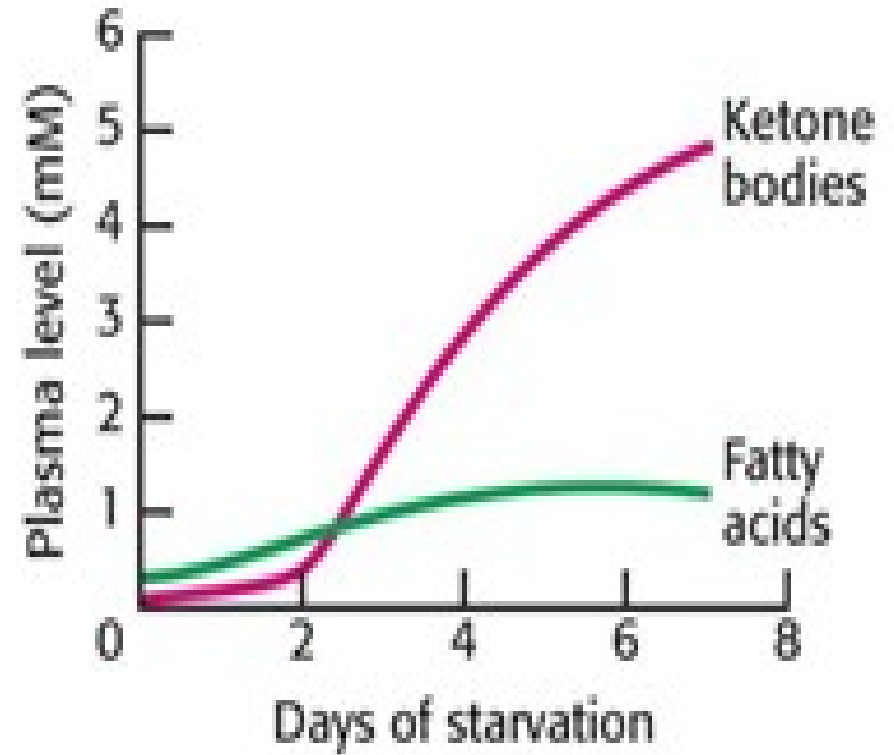
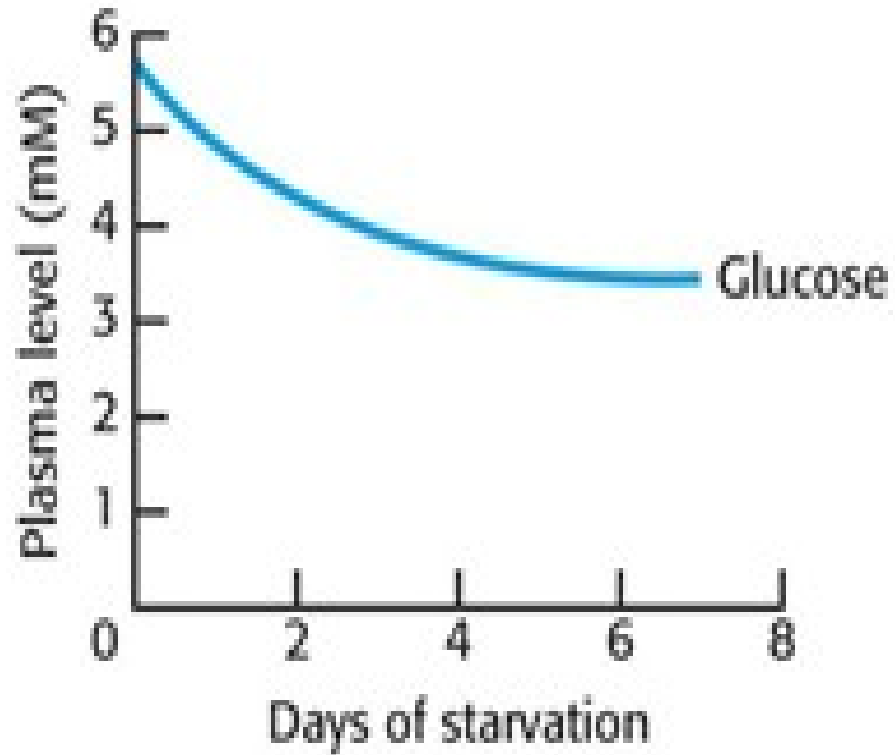




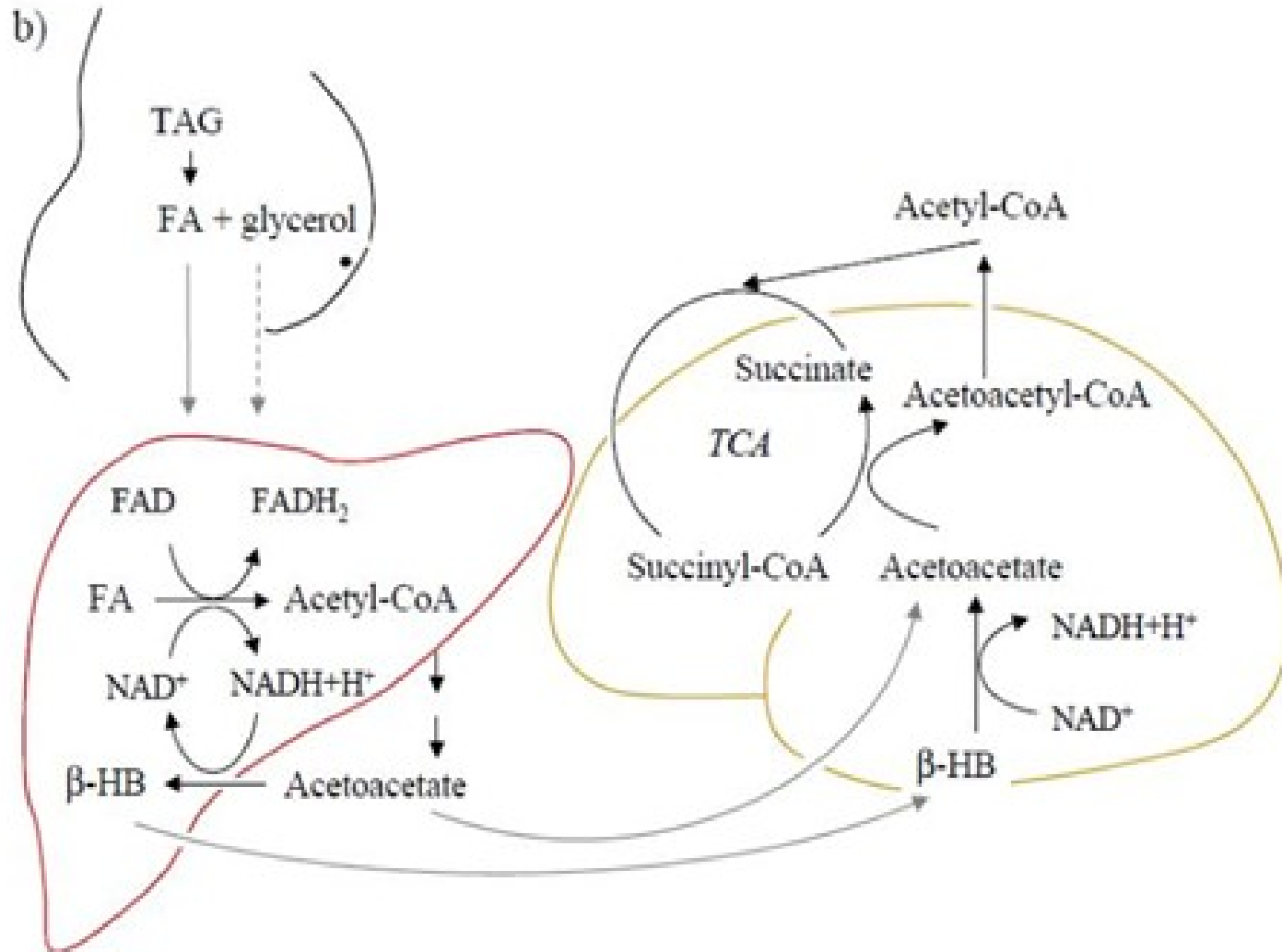
lactate, ketone bodies, and amino acids,

- Prolonged fasting or ketogenic diet
 - increases the blood level of ketone bodies and results in enhanced use by the heart.
- On the other hand, **cardiac extraction and oxidation of lactate becomes predominant** during
 - as skeletal muscle lactate production increases
- In isolated perfused hearts, the addition of lactate or ketone bodies in the perfusate significantly
 - reduces the oxidation of glucose and fatty acids.

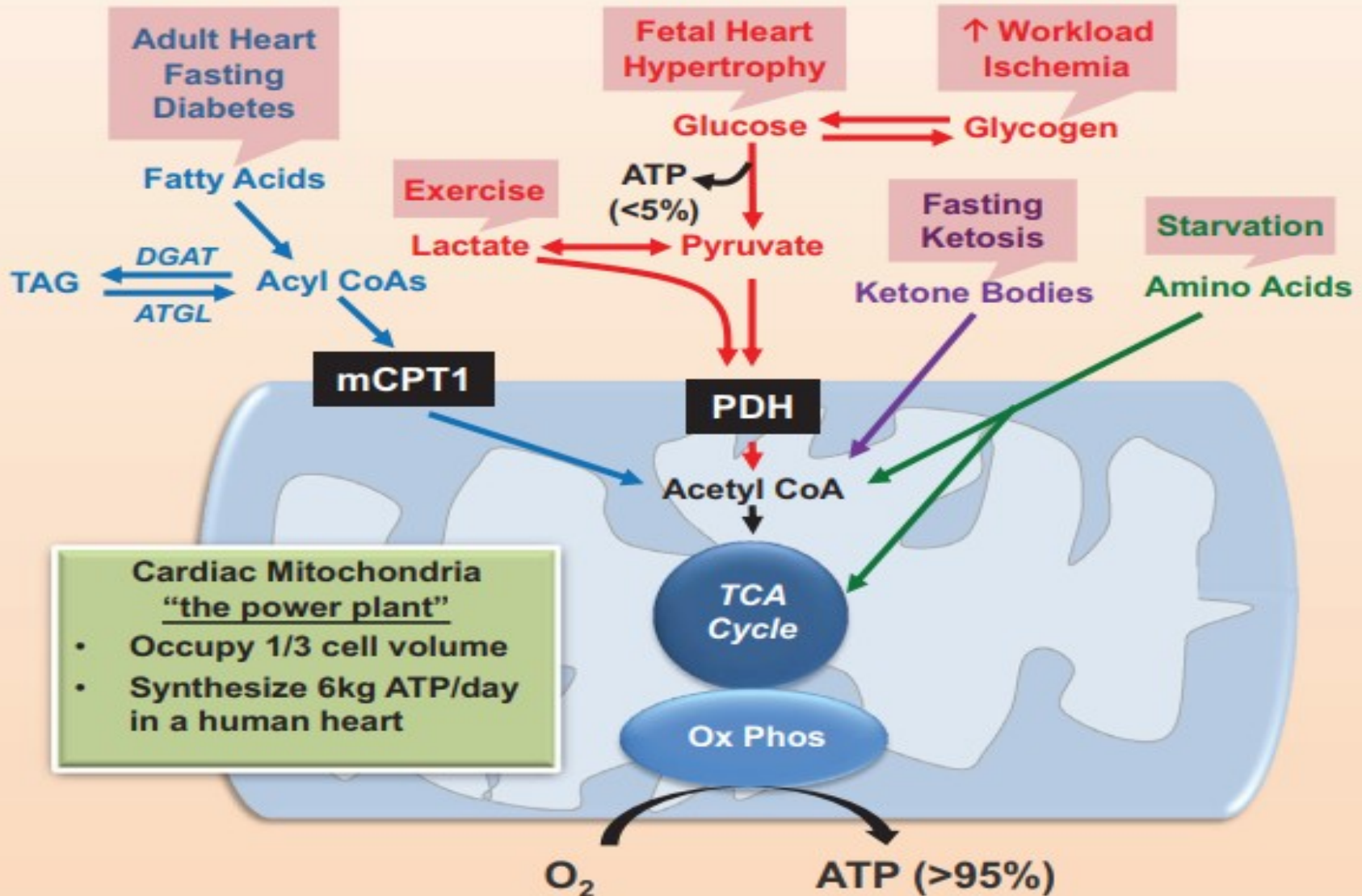
Starvation



Production of ketone bodies



Overview of the metabolic network





Intermediates of Glucose Metabolism

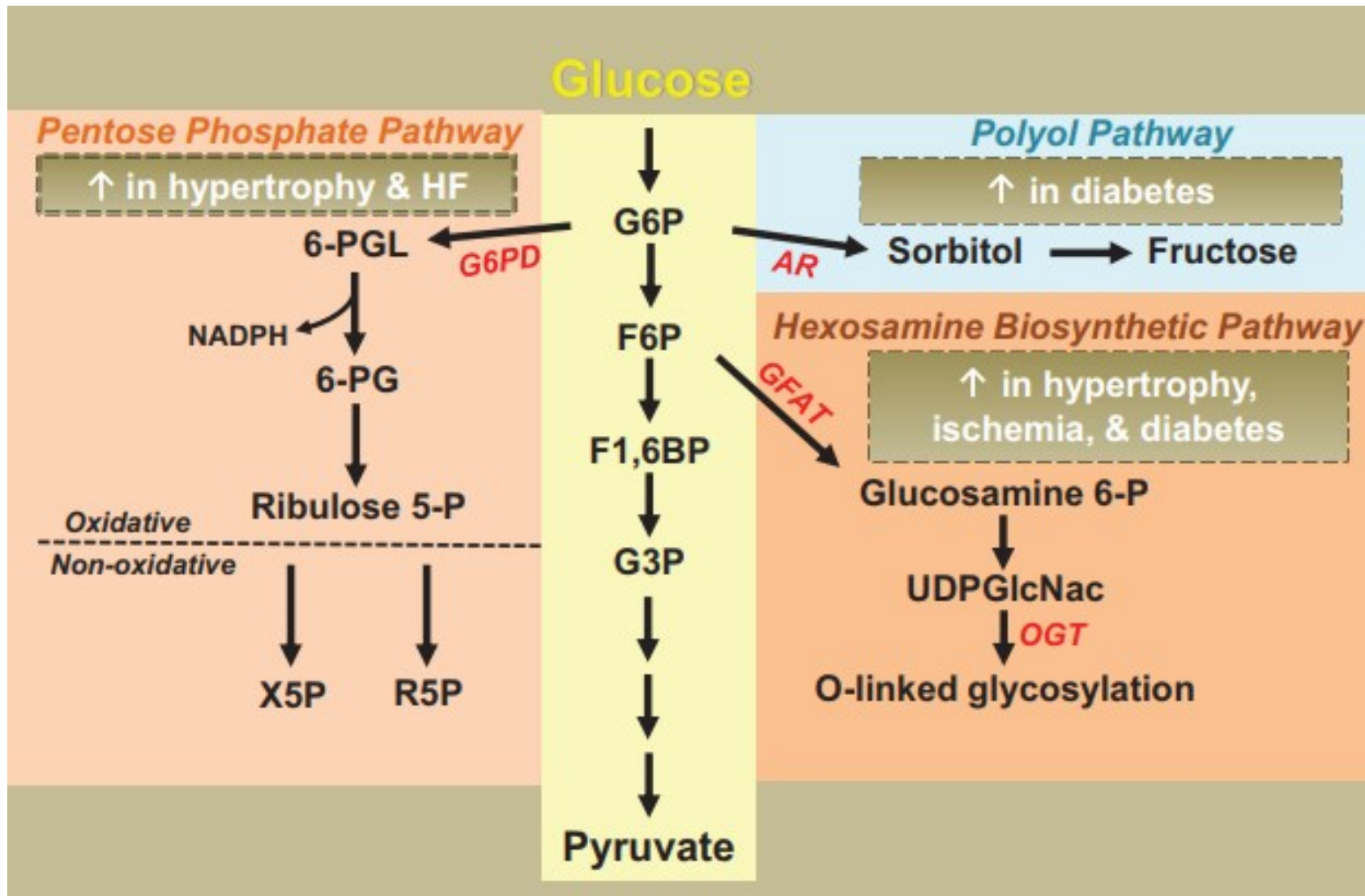
- Although
 - glucose $\rightarrow$ pyruvate, ATP production,
 - glycolytic intermediates participate in several additional pathways .
- These pathways are of **biological significance** in the **heart despite the small fluxes**.
- G6P produced by the hexokinase enters the PPP, yielding
 - NADPH during the oxidative phase and
 - 5-carbon sugars in the subsequent non-oxidative phase.

Intermediates of Glucose Metabolism...

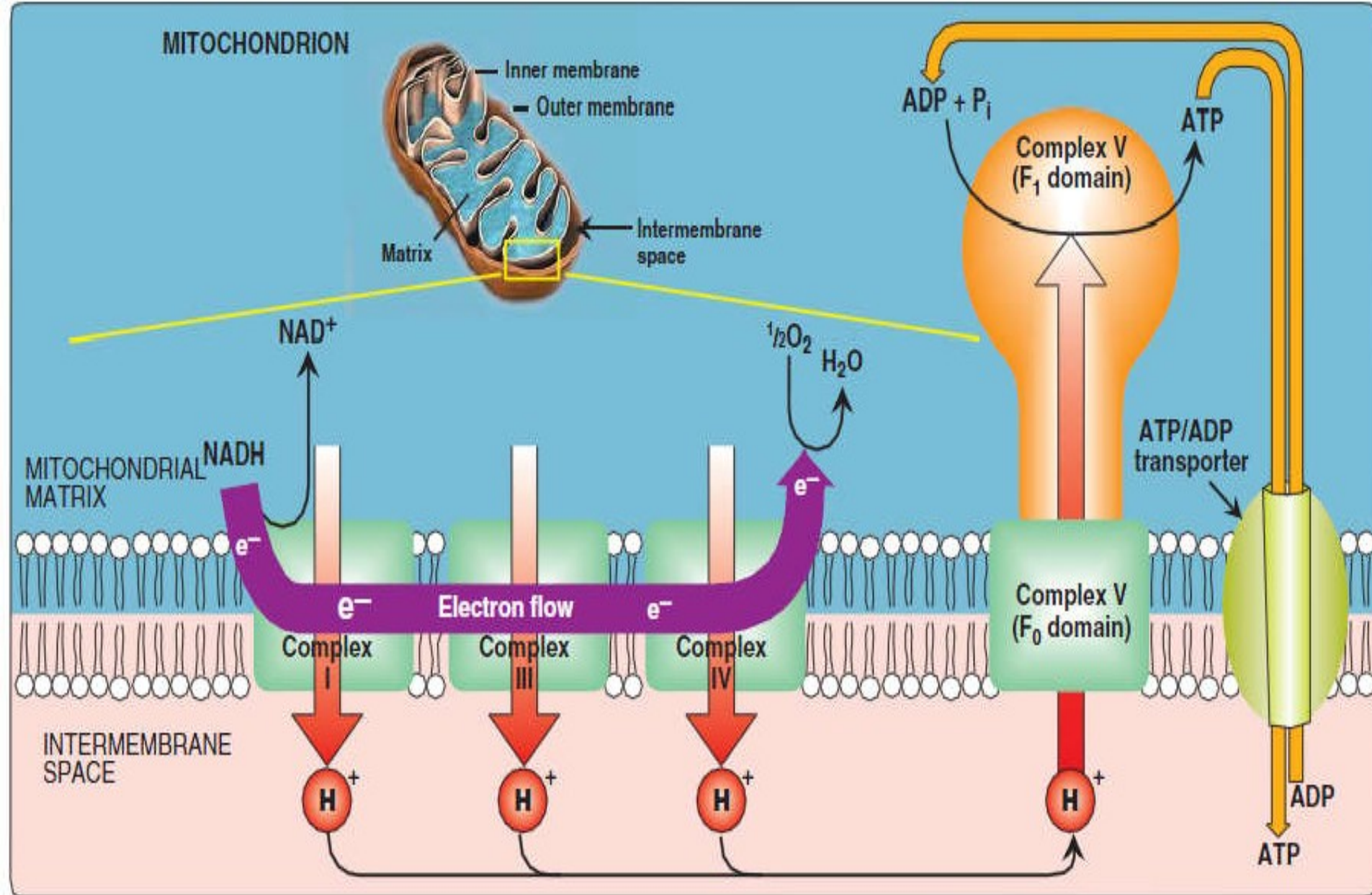


- NADPH important for **antioxidant defense** because
 - NADPH is required for maintaining the level of **reduced glutathione**.
 - Deficiency of G6PD, **exacerbates ischemia–reperfusion injury**
- Ribose 5-phosphate becomes a substrate for nucleotide or nucleic acid synthesis,

Intermediates of Glucose Metabolism....



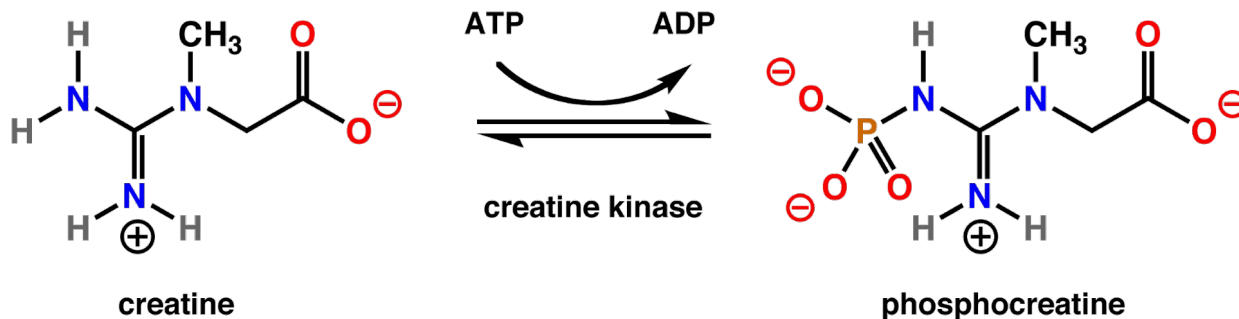
Generation of ATP via the ETC



Energy Transfer



- The predominant mechanism of energy transfer is –the **creatine kinase (CK) system**
- The forward reaction occurs within the **mitochondrial intermembrane space**.
- The resultant **PCr** is relatively less polar and can store to high levels in the cytosol.
- **PCr** available for rapid renewal of ATP at times of high need.
- This reverse reaction is catalyzed by cytosolic isoform of CK



Regulation



- Regulation of fatty acid and carbohydrate metabolism is described by the phenomenon known as

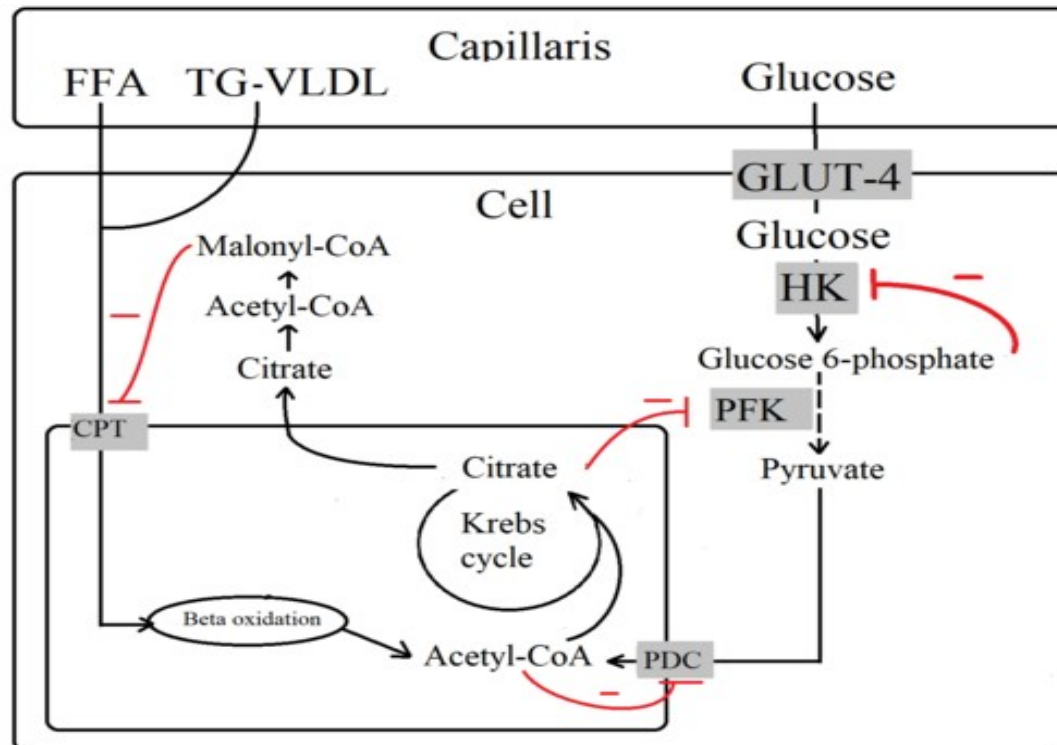
–Randle Cycle

- Whereby **increased fatty** acid utilization results in
 - increased **acetyl CoA production**, thereby inhibiting PDH activity, and
 - **increases cytoplasmic citrate**, which in turn inhibits glycolysis

Regulation



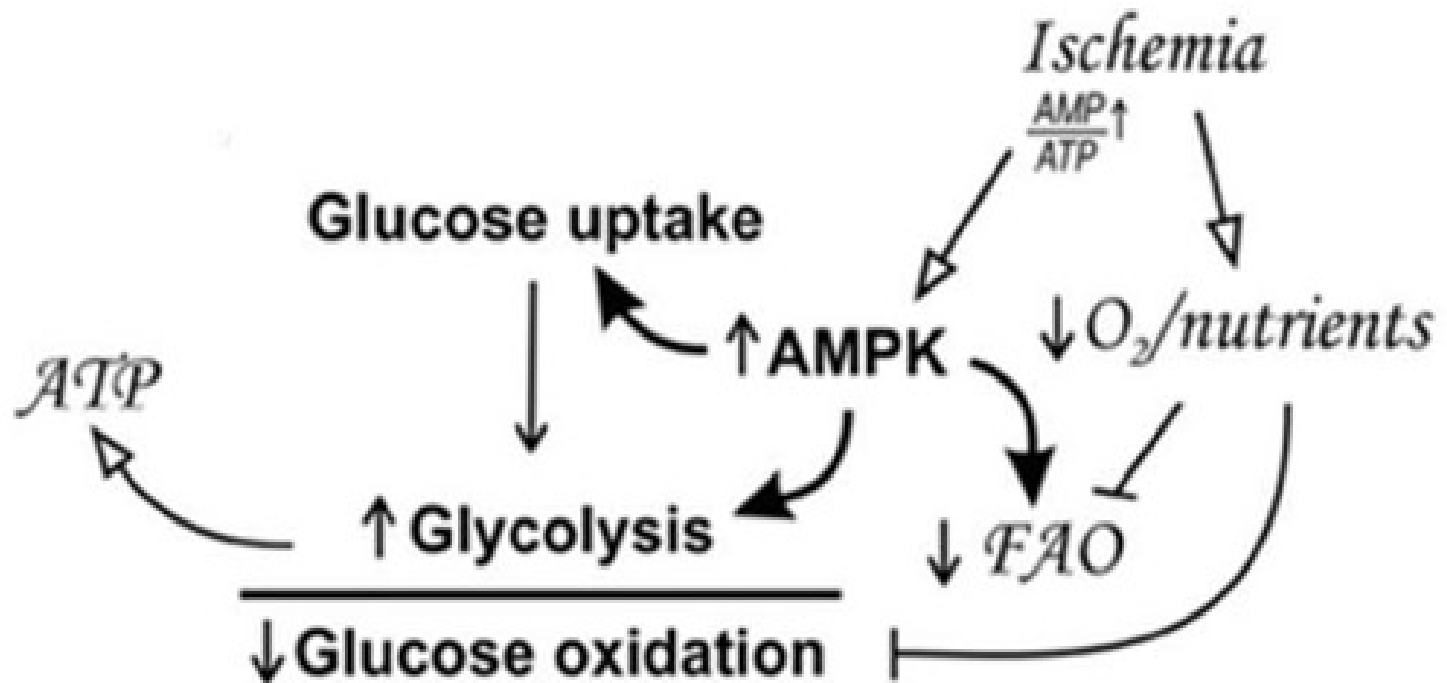
- Conversely, when **carbohydrate metabolism is increased**,
 - citrate from the TCA cycle** is exported to the cytoplasm where it is converted to **acetyl CoA** → **malonyl CoA** → inhibition of **CPT-1 activity**
 - thereby reducing fatty acid metabolism.



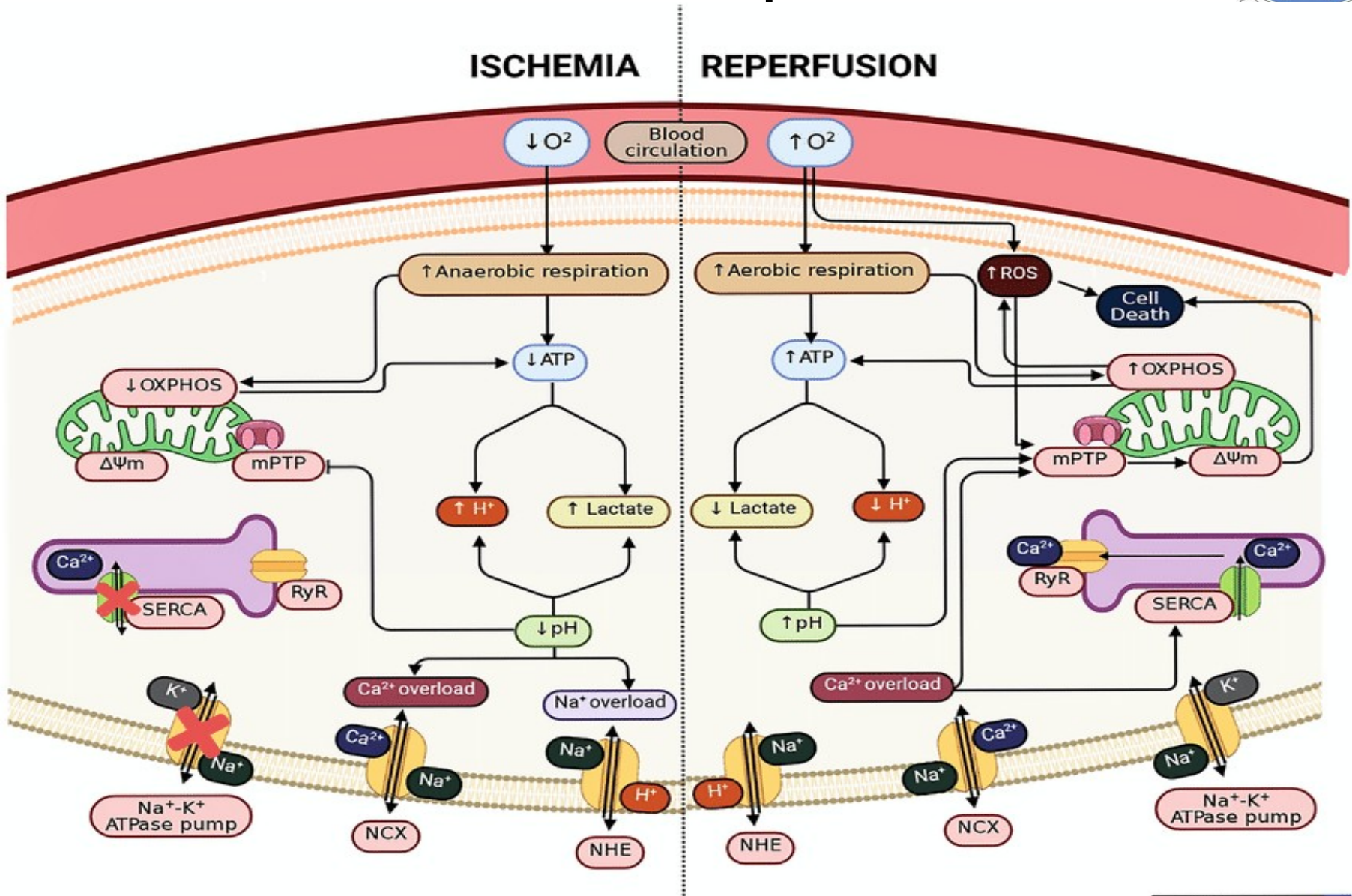
How does ischemia alter metabolic profile?



- Ischemia results in
 - Decrease of O_2 and nutrients, which inhibit fatty acid oxidation
 - Increase in AMP/ATP ratio, which activates AMPK



Ischemic vs. Reperfusion



Biomarkers of Cardiac Injury



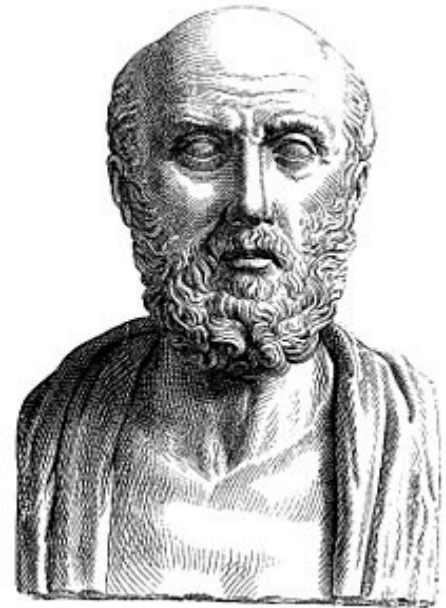
What are biomarkers?

- A biomarker is a substance used as an indicator of the **biologic state**.
- Characteristic **objectively measured** and evaluated as an indicator of
 - normal biologic processes
 - pathogenic processes or
 - pharmacologic responses to a therapeutic intervention.



Biomarkers of Cardiac Injury

- In his teachings, **Hippocrates** inferred the importance of dynamic **circulatory factors** in mediating illness.
- The search for these "intense and strong juices" has led to the discovery of biomarkers in CV medicine.
- Today, in patients with known or suspected CVD, biomarkers are reported to be associated with the risk, presence, and severity of disease.



Cardiac biomarkers(CBM)



- CBM- Are **protein molecules** released into the blood stream from damaged heart muscle with chxs $\updownarrow$ pattern
- **Cardiac marker:** a **clinical laboratory** test useful in cardiac disease, most commonly for detecting myocardial injury.
- **Characteristics of ideal cardiac marker**
 - High cardiac specificity
 - Easy diagnosis - sensitivity
 - Marker should play a designed role in the treatment and management of clinical subject.

Classification of Cardiac Biomarkers

1. Biomarkers of myocardial injury

- Creatine kinase-MB
- Myoglobin
- Cardiac troponins

2. Biomarkers of hemodynamic stress

- Natriuretic peptides

3. Inflammatory and prognostic biomarkers

- C-reactive protein
- Myeloperoxidase
- Homocysteine

Biomarkers of Myocardial Injury

Myocardial infarction is due to:
Occlusion of coronary arteries



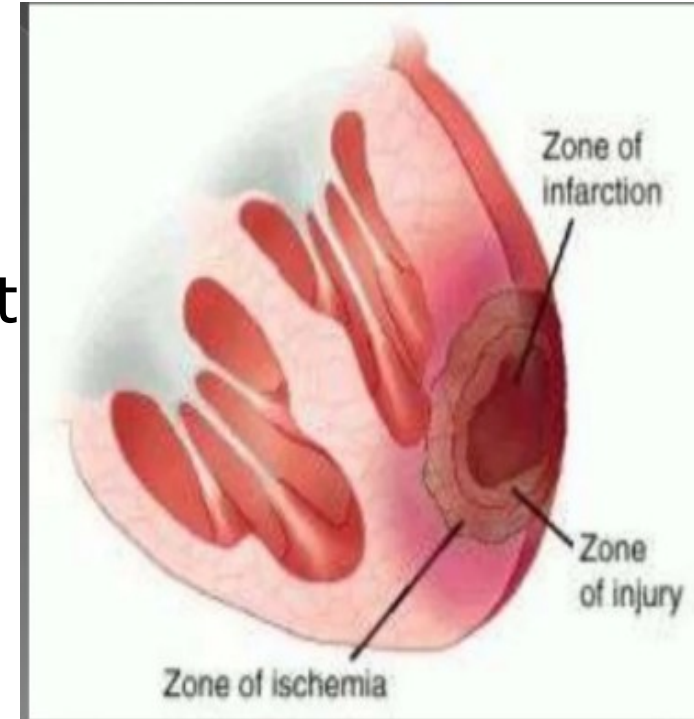
restricted blood supply (oxygen) to heart tissue (**ischemia**)



damage to heart tissue (**infarction**)



release of enzymes and other proteins into the blood (**Creatine kinase-MB, Myoglobin, Cardiac troponins**)



Creatine kinase



- Is an enzyme expressed in a number of tissues
- It catalyzes the conversion of creatine to phosphocreatine
- Consists of two subunits: B (brain type) or M (muscle type)
 - Making three different isozymes

Multiple Forms of Creatine Kinase (CK):

Form	Description
CK-MM	predominates in the skeletal muscles.
CK-MB	predominates in the cardiac muscle.
CK-BB	predominates in the brain, smooth muscles, and lungs.

Creatine kinase-MB



- High specificity for cardiac tissues
- Begins to rise 4-6 hours after onset of infarction
 - Peaks at about 12 hours
- Returns to baseline at 24-36 hours
 - Can be used to indicate **early re-infarction** if the level normalizes and the increases again

Myoglobin



- Found in cardiac and skeletal muscles
- Single polypeptide with one oxygen binding site
- It contains a heme prosthetic group that can reversibly bind to oxygen
- The body uses it as **an oxygen storage protein in muscle**
- Bind and release oxygen depending on the oxygen conc. in the cell → primary function to supply oxygen to myocytes
- It is the reason for the red color of the muscle

Myoglobin Cont.



- It is sensitive index of myocardial damage
- Surges often occur earlier than Tn & CK
 - ✓ Released from damaged tissues within 1 hour
- It is not specific
- Conditions that increase myoglobin
 - Acute myocardial infarction
 - Skeletal muscle damage
 - Renal failure

Troponin

- A complex of three regulatory proteins that is integral to muscle contraction in skeletal & cardiac muscle
 - Involved in the interaction between actin and myosin for contraction.
- It has three subunits: TnC, TnT & TnI
 - Troponin-C has calcium binding ability and has no diagnostic value
 - Troponin-T binds the troponin-tropomyosin complex
 - Troponin-I is an inhibitory protein

Troponin Cont.

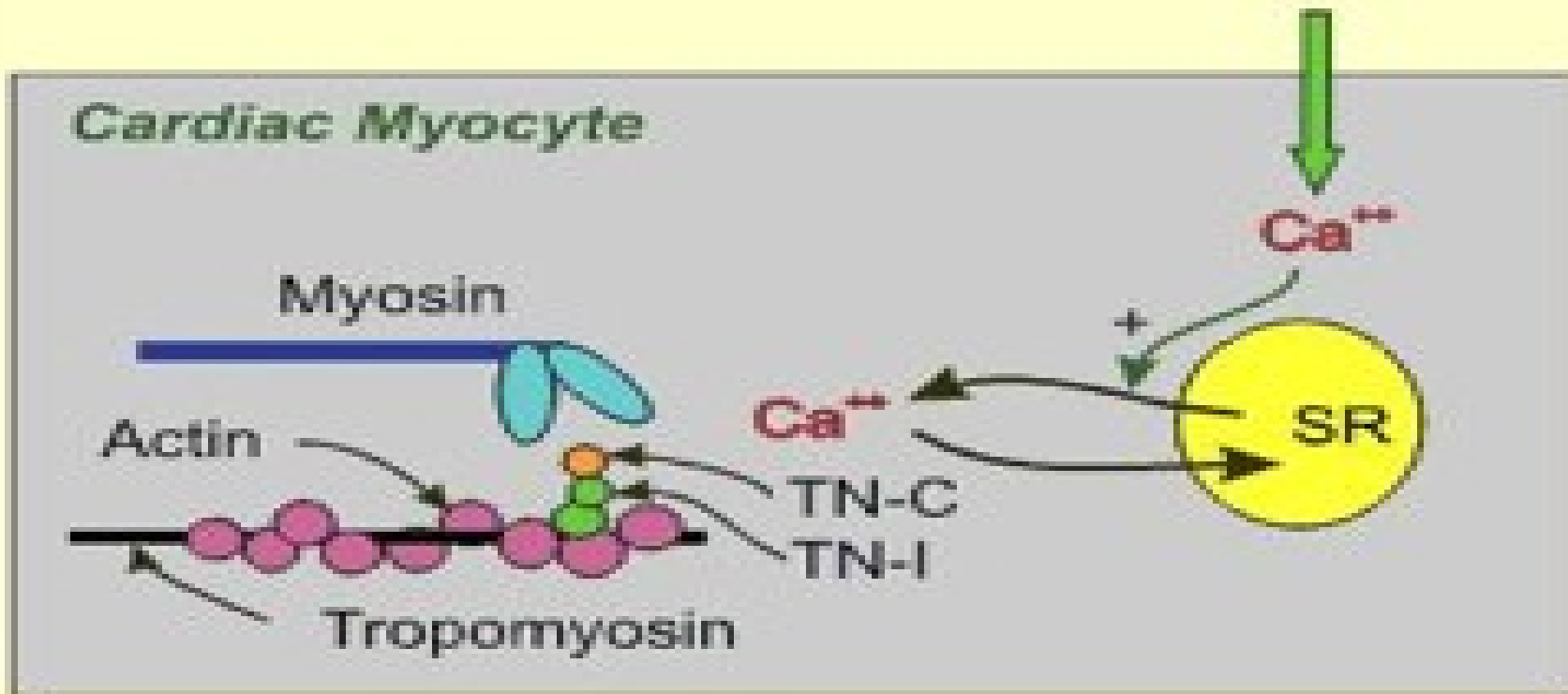


Figure 2. Cardiac myofilaments. Myosin (thick filament) contains two heads having ATPase activity. Thin filament is made up of actin, tropomyosin, and troponin (TN). TN-C binds Ca^{++} released by the sarcoplasmic reticulum (SR). TN-I inhibits actin-myosin binding until Ca^{++} binds to TN-C.

Troponin C



- 2 isoforms (cardiac and slow-twitch muscle)
 - ❖ Homology between isoforms limits cardiac specificity and diagnostic usefulness

- Binds calcium to initiate muscle contraction

- Calcium binds cTnC → cTnI displaced → tropomyosin conformational change (no longer interferes with myosin/actin) → muscle contraction occurs

Troponin I



- Unique to heart cardiac protein molecular weight of 24 kilo Daltons
 - Healthy animals may have low, but detectable levels
 - Time to initial elevation = 3-8 hours
 - Time to peak elevation = 12-24 hours
 - Time to return to baseline = 5-10 days
- cTnI has additional amino acid residues **on its N-terminal that are not exist on the skeletal form**
- Half life 2-4 hours & serum increase within 2-8 hours
- Troponin I
 - 3 isoforms (2 skeletal, ***1 cardiac***)



Troponin T

- Cardiac troponin T (cTnT) is present in **fetal skeletal muscle**.
- In healthy adult skeletal muscle cTnT is absent.
- Has similar half life and early serum increases as that of cTnI
- Troponin T
 - Multiple isoforms in skeletal tissue
 - 4 isoforms in human cardiac tissue (1 adult, 3 fetal tissue)
 - Fetal isoforms may be re-expressed during heart failure or in damaged skeletal muscle**
 - Attaches the troponin complex to tropomyosin and actin

Troponin levels

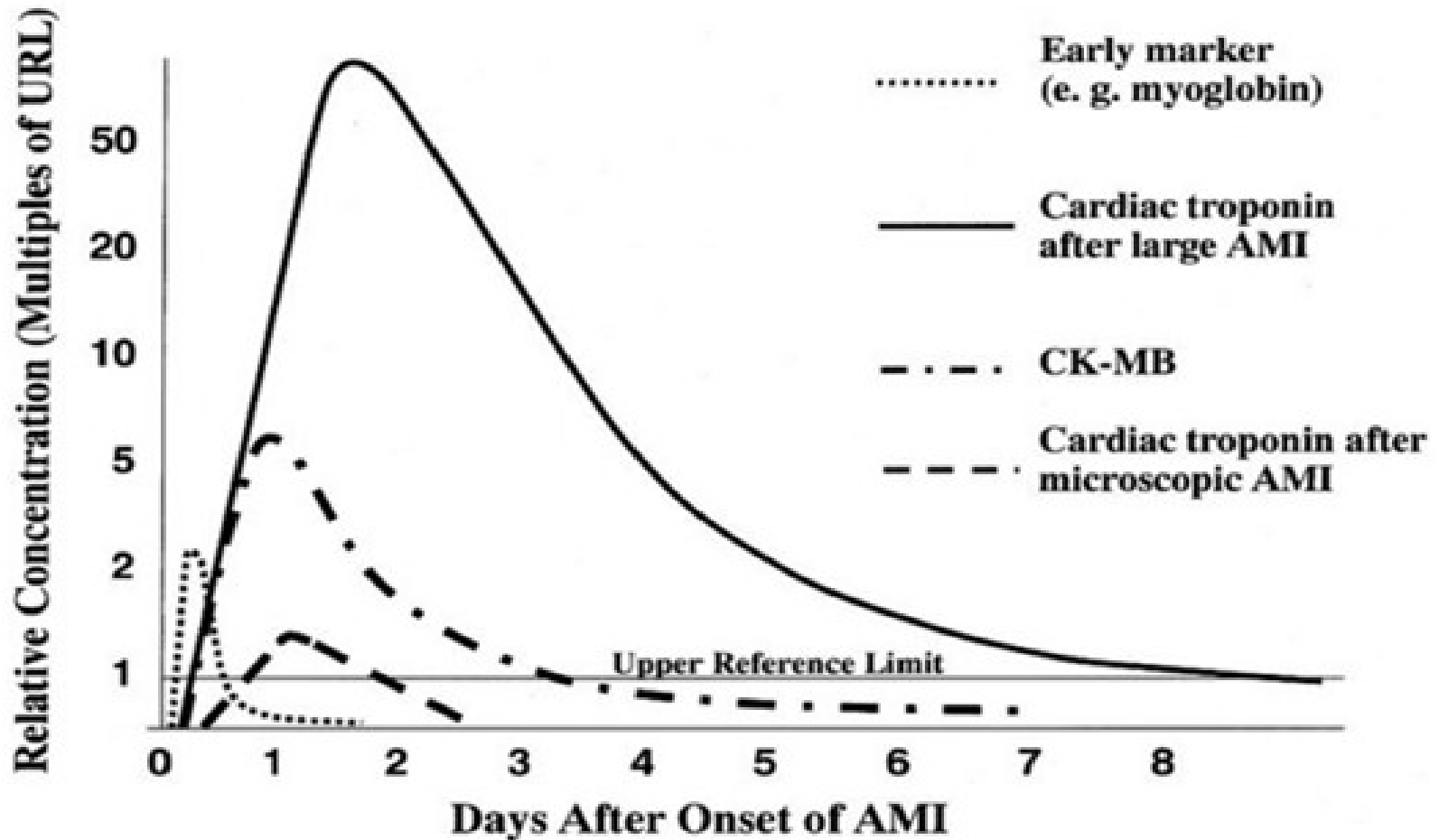
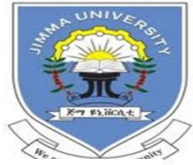


- Less than 5% in cytosol
- Troponin levels begin to rise 2-3 hours after onset of **myocardial injury**
- Elevations of TnI & TnT can persist for up to 10 days after MI.

NB:

- CK-MB returns to baseline by 48 hours

Summary of MI markers



Natriuretic peptides

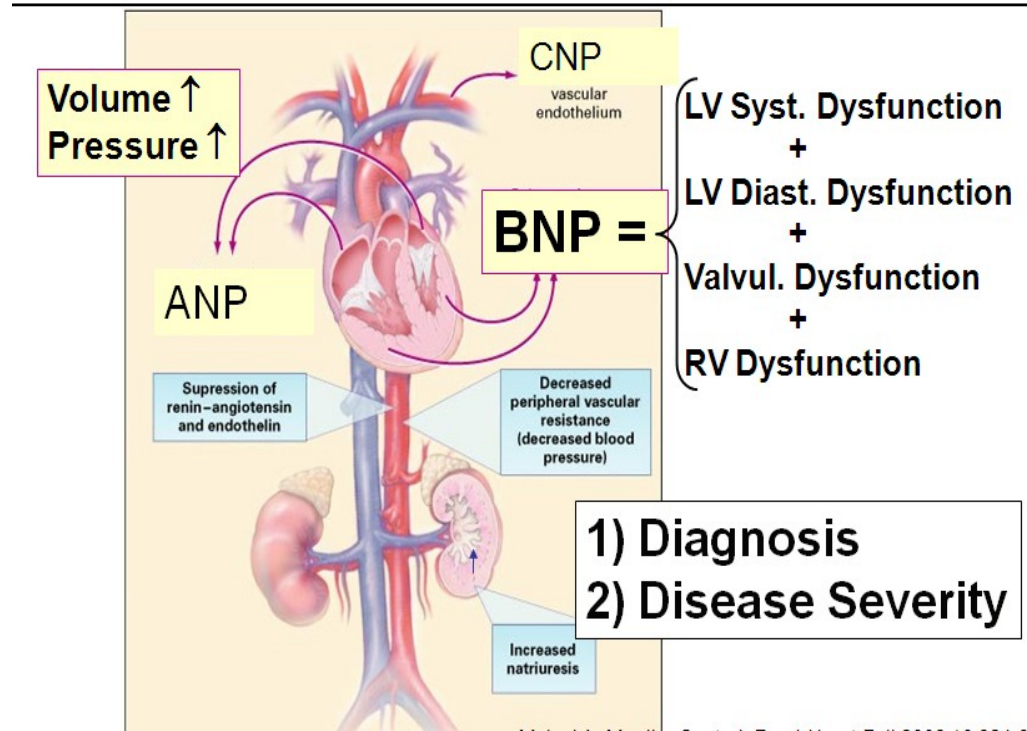


- NPs are group of structurally similar but **genetically distinct** neurohormones which acts to maintain **CV homeostasis**.
- NPs family includes
 - ANP**: **atrial natriuretic peptide** (28 aa)
 - predominantly synthesized in the atria and released in response to **atrial distension**.
 - BNP**: **B-type natriuretic peptide** (32 aa)
 - mainly produced and secreted by **ventricular myocytes** following volume overload →ventricular wall stretch .
 - CNP**: **C-type natriuretic peptide** (22 aa)
 - Is found predominantly in the brain and also synthesized by vascular endothelial cells.

Natriuretic peptides cont.

FIGURE 2

NP: Marker of Cardiac Stress



Maisel A, Mueller C, et al. Eur J Heart Fail 2008;10:824-39

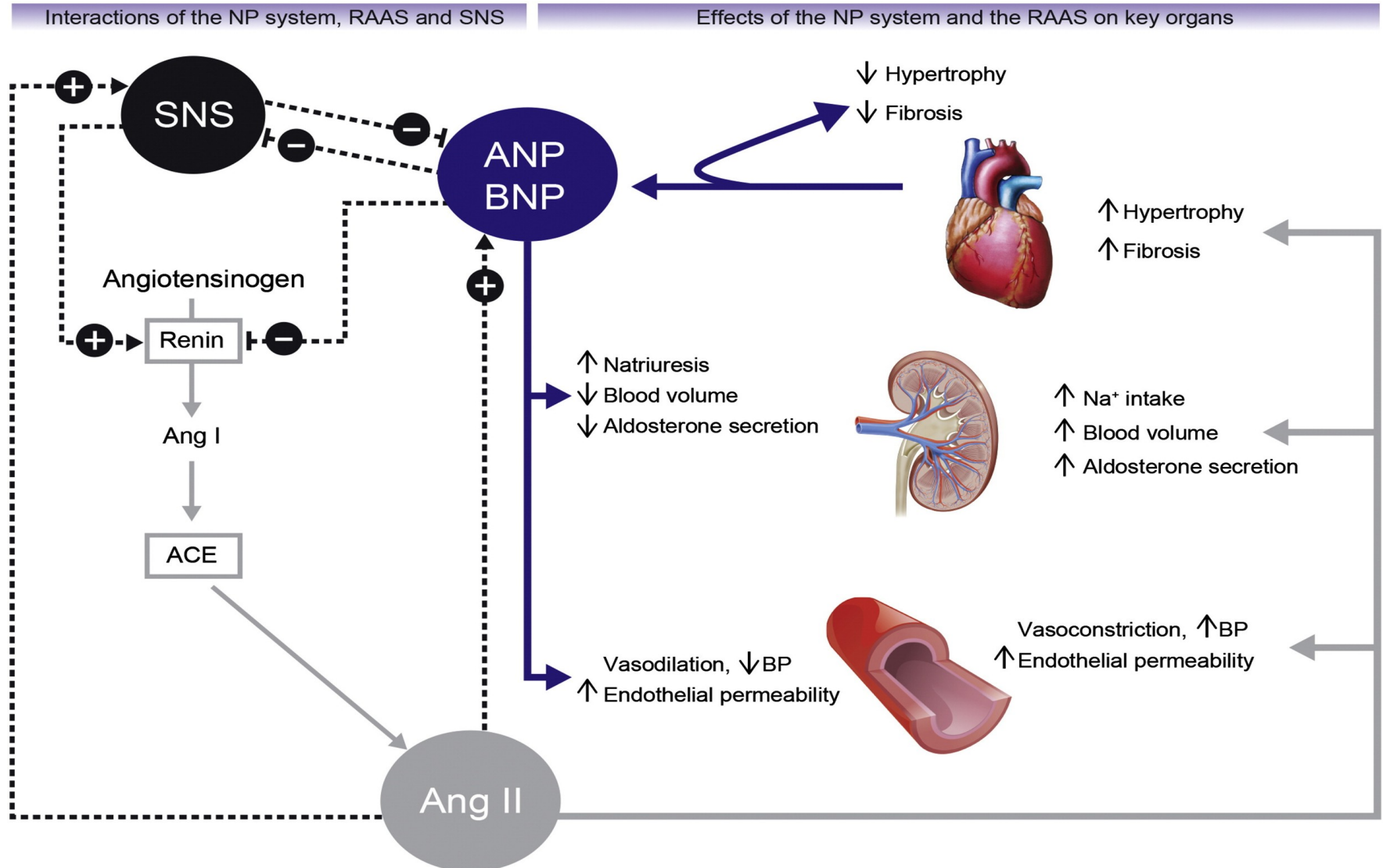
- In contrast to ANP, the circulating physiological levels of BNP are generally very low and
 - only become more notable in **pathological disease states.**

Comparison different families of NPs



NP	ANP	BNP	CNP
Structure	28 amino acid peptide	32 aa peptide	22 aa peptide
Site of synthesis	Atria	Ventricles	Vascular endothelial cells
Trigger(s)	Atrial distension	Volume overload (→ventr. wall stretch)	Cytokines e.g. IL-1, TNF
Proposed main role	Physiological	Cardiac stress hormone	?
Receptor	NPR-A	NPR-A	NPR-B
Receptor coupling mechanism	↑Guanylate cyclase/cGMP	↑Guanylate cyclase/cGMP	↑Guanylate cyclase/cGMP
Main physiological effects of NP	Natriuresis, diuresis Vasodilation, ↑Renal blood flow & GFR ↑Myocardial relaxation	Natriuresis, diuresis Vasodilation, ↑Renal blood flow & GFR ↑Myocardial relaxation Lipid mobilization,	Vasodilation Antihypertrophic and antifibrotic Anti-inflammatory Antithrombotic

Natriuretic peptides cont.



C-reactive protein

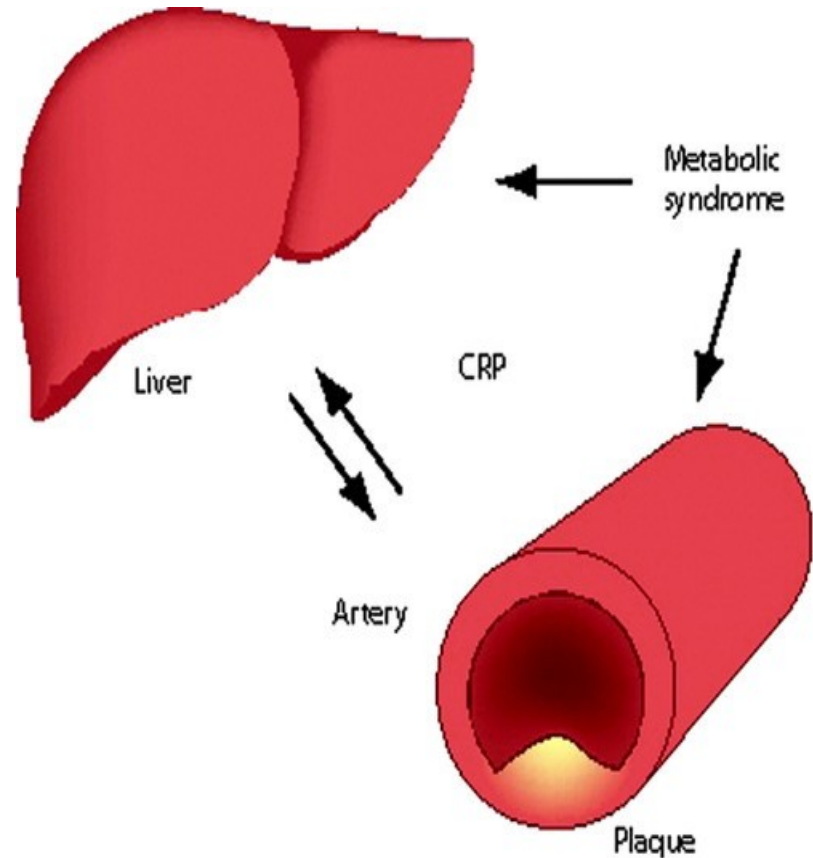


- CRP is an acute-phase protein produced by the liver
- Pentameric structure consisting of five 23 kDa identical subunits
- Plasma levels can increase rapidly to 10,000x levels
- CRP useful **marker for increased cardiovascular risk.**
- This is an attractive concept because **atherosclerosis** is a disease characterized by **chronic arterial inflammation** and
 - suggests the possibility that subclinical states of atherosclerosis.

C-reactive protein Cont.



- Mechanism of CRP-mediated atherogenesis: once ligand bound CRP can
 - Activate classical complement pathway
 - Stimulate phagocytosis
 - Bind to Ig receptors
 - Endothelial dysfunction via increase NO synthesis
 - Increase LDL deposition in plaque by CRP- stimulated macrophages



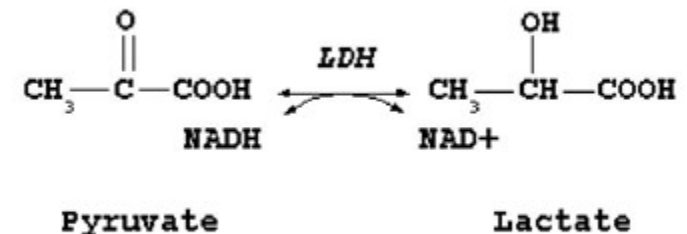
Less Commonly Used Cardiac Markers

- Lactate dehydrogenase system
 - were used widely in the past for diagnosis of myocardial infarction
 - exists in five isoenzyme forms

LDH ISOENZYMES

LDH exists in 5 forms (isoenzymes), which differ slightly in structure. • All of these isoenzymes can be measured in the blood, and can be separated by electrophoresis.

LDH Isoenzyme	Tissues
LDH-1	is found primarily in heart muscle and red blood cells.
LDH-2	is concentrated in white blood cells.
LDH-3	is highest in the lung
LDH-4	is highest in the kidney, placenta, and pancreas
LDH-5	is highest in the liver and in skeletal muscle



LD Enzymes Cont.



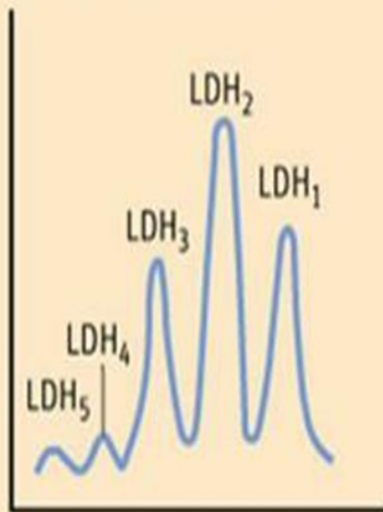
- In particular, these enzymes and isoenzymes carry out their metabolic functions within specific cells, and are released into body fluids when those cells become diseased.
- Thus an increase in LD isoenzyme activity can indicate leakage from cell due to cellular injury.
- The level of isoenzyme LD1 compared with LD2 has been used to detect an AMI because of the high concentration of LD in cardiac muscle fibres.
- LD isoenzymes begin to leak out of dying heart muscle cells and are detectable in the serum by 12 to 24 hours following a heart attack

LDL Isoenzymes

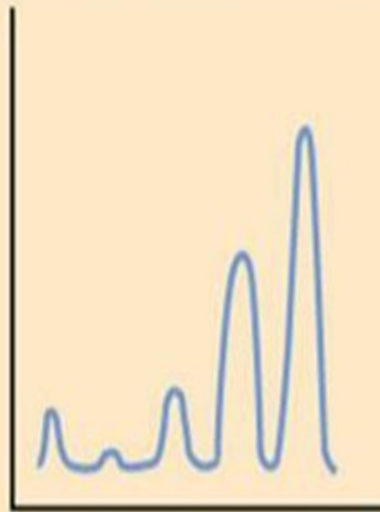
Densitometric patterns of LDH isozymes in normal and patient serum



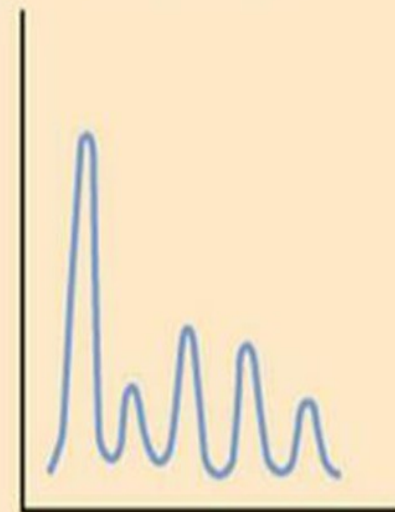
A Normal serum



B Acute myocardial infarction



C Acute hepatitis





Reading Assignment

- Myeloperoxidase
- Homocysteine



Thanks!!